SEUNGPIL LEE

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Personal Statement
Master's student who wants to understand how intelligence works, guided by the research philosophy "from humans to AI, and from AI back to humans." My previous research mainly explored how language models "think." My current interest is metacognition. I am also interested in a broad range of fields, including science and technology studies (STS) and literature.

EDUCATION

Gwangju Institute of Science and Technology (GIST) *March 2025 – present*
Master's Program, Major in AI Convergence

Gwangju Institute of Science and Technology (GIST) *March 2018 – December 2024*
B.S. in Electrical Engineering and Computer Science GPA: 3.66/4.50
Minor in Mathematics, Minor in Literature

UC Berkeley *June 2019 – August 2019*
Berkeley Summer Session Program GPA: 4.00/4.00

PUBLICATIONS

Journal Articles

- **Seungpil Lee***, Woochang Sim*, Donghyeon Shin*, Sanha Hwang, Wongyu Seo, Jiwon Park, Seokki Lee, Sejin Kim, and Sundong Kim. "Reasoning Abilities of Large Language Models: In-Depth Analysis on the Abstraction and Reasoning Corpus." *ACM Transactions on Intelligent Systems and Technology* (**ACM TIST**), 2025.
- Sanha Hwang, **Seungpil Lee**, Sejin Kim, and Sundong Kim. "Solution Augmentation for ARC-AGI Problems Using GFlowNet: A Probabilistic Exploration Approach." *Transactions on Machine Learning Research* (**TMLR**), 2025.

Conference Papers

- Heejun Kim*, **Seungpil Lee***, Jewon Yeom, Jaewon Sok, Seonghyeon Park, Jeongjae Park, Taesup Kim, and Sundong Kim. "From Noise to Diversity: Random Embedding Injection in LLM Reasoning." *Conference on Neural Information Processing Systems* (**NeurIPS**), 2026. (Earlier version at ICML Mechanistic Interpretability Workshop 2026, Spotlight)
- **Seungpil Lee**, Donghyeon Shin, Yunjeong Lee, and Sundong Kim. "Can Large Language Models Develop Gambling Addiction?" *Conference on Neural Information Processing Systems* (**NeurIPS**), 2026.
- **Seungpil Lee***, Donghyeon Shin*, Klea Lena Kovacec, and Sundong Kim. "From Generation to Selection: Findings of Converting Analogical Problem-Solving into Multiple-Choice Questions." *Findings of EMNLP* (**EMNLP Findings**), 2024.
- Hosung Lee*, Sejin Kim*, **Seungpil Lee**, Sanha Hwang, Jihwan Lee, Byung-Jun Lee, and Sundong Kim. "ARCLE: The Abstract and Reasoning Corpus Learning Environment for Reinforcement Learning." *Conference on Lifelong Learning Agents* (**CoLLAs**), 2024.

* indicates equal contribution

SKILLS

Programming	C, C++, Java, JavaScript, Python
Tools	LaTeX, Spring Boot, PyTorch, Hugging Face
Language	Korean (Native), English (Intermediate)

EXPERIENCE

LG AI Research, Seoul *September 2026 – present*
Research Intern

- Improving the coding capability of K-Exaone, a Korean sovereign foundation model

Microsoft Research Asia (MSRA), Shanghai *February 2026 – August 2026*
Research Intern, Collaborative Research Program

- Developed test-time skill optimization for LLM agents, letting them acquire and reuse task skills at inference time without parameter retraining
- Extended the Agent-Lightning framework with memory generation so agents carry learned context across long-horizon tasks
- Bridged mechanistic analysis of reasoning failures with memory-based mitigations

DataScience Lab, GIST *September 2023 – present*
Undergraduate Intern → Master's Student

- LLM skill discovery and evaluation with cognitive science tools (Language of Thought Hypothesis)
- LLM alignment using RLVR methods; Mechanistic interpretability with Sparse Autoencoders

Teaching Assistant, Computer Networking *September 2023 – December 2025*

- Designed assignments connecting theory to implementation; SDN demo with Ryu and Mininet

Development Team, Republic of Korea Navy *January 2022 – September 2023*

- Mandatory military service; back-end developer for internal app and web services

BioComputing Lab, GIST *June 2020 – January 2021*
Undergraduate Intern

- Developed Spiking Neural Network methods with synthetic gradients

AWARDS & FUNDING

National Research Foundation of Korea (NRF) Funding *Sep 2025 – Sep 2026*
Research funding for outstanding master's students

Best Undergraduate Thesis Award, GIST EECS *February 2025*
Awarded for graduation thesis on MC-LARC benchmark

Korean Government Scholarship, GIST College *March 2018 – August 2024*
Full scholarship for students at GIST

Scholarship for Summer Session Abroad *June – August 2019*
UC Berkeley Summer Session scholarship

Navy AI Competition *August 2023*
Award in drone object detection competition

K-StartUp Finals *November 2023*
Advanced to finals as ML researcher and back-end developer